SENSE SIGHT

A PROJECT REPORT

Submitted by

Krishna (23BAI10621)

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VIT BHOPAL UNIVERSITY

KOTHRIKALAN, SEHORE
MADHYA PRADESH - 466114

LIST OF ABBREVIATIONS

LiDAR: Light Detection and Ranging
YOLO: You Only Look Once
SSD: Single Shot Multibox Detector
SOC: System on Chip

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ABSTRACT

Vision impairment affects millions of individuals globally, posing significant challenges to their independence, mobility, and safety. Navigating public spaces, avoiding obstacles, and engaging in social interactions often require external assistance or specialized tools. Traditional aids, such as white canes, provide basic support but are inadequate for detecting dynamic obstacles and ensuring comprehensive safety in complex environments.

The Sense-Sight project offers an innovative technological solution designed to enhance the spatial awareness and safety of visually impaired users. By integrating state-of-the-art object detection algorithms, proximity sensors, and real-time feedback mechanisms, this system provides users with a reliable and practical assistive tool. Key features include the use of YOLO (You Only Look Once) for precise and efficient object detection, LiDAR/ultrasonic sensors for proximity analysis, and audio or haptic feedback to alert users of nearby obstacles.

This wearable system is lightweight and portable, making it suitable for everyday use. It aims to bridge the gap between existing assistive technologies and the specific needs of visually impaired individuals, particularly in dynamic and crowded settings. The project's methodology emphasizes real-time processing, user-friendly design, and the potential for integration with modern wearable devices. By addressing limitations in current solutions, Sense-Sight aspires to significantly improve the quality of life and independence of its users.

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CHAPTER 1: PROJECT DESCRIPTION AND OUTLINE

1.1 Introduction

The sense of sight is one of the most critical human sensory systems, allowing us to perceive, interpret, and respond to our surroundings. Vision provides the majority of information we use to interact with the world, from recognizing faces and objects to understanding spatial relationships and navigating safely. Inspired by the importance of vision, this project seeks to develop a system that replicates its essential functions using advanced technologies.

The proposed system combines object detection with auditory feedback to create a practical and assistive tool for visually impaired individuals or users who need situational awareness. By leveraging the power of the YOLO (You Only Look Once) object detection model and real-time video processing, the system identifies objects in a live video feed and notifies the user when a target object is detected. This approach integrates computer vision and speech technology to enhance accessibility, independence, and safety for users in various scenarios.

1.2 Motivation for the Work

The motivation for this project stems from the challenges faced by individuals with visual impairments and the growing demand for assistive technologies that enhance quality of life. Visual impairments can significantly impact an individual's ability to perform daily tasks, navigate unfamiliar environments, and maintain safety. While many assistive technologies exist, few offer a real-time, customizable, and intuitive solution that combines object detection with auditory feedback.

Additionally, the rapid advancements in computer vision and deep learning provide an opportunity to design innovative systems that replicate human capabilities, including vision. By utilizing these advancements, this project aims to create a robust and user-friendly solution that bridges the gap between need and technology, offering practical applications for both personal and industrial use cases.

1.3 About the Project

This project implements an intelligent object detection system that uses the YOLO algorithm for real-time video processing. The system's core functionality involves detecting objects in the environment and providing auditory feedback when specified target objects are identified. The user can customize the list of target objects by editing a text file, making the system adaptable to different needs and environments.

Key components of the project include:

- YOLO Object Detection: A deep learning-based algorithm that is optimized for speed and accuracy, capable of detecting multiple objects in a single pass.
- Real-Time Video Processing: Capturing and analyzing frames from a live video feed to provide instant feedback.
- Auditory Notifications: A text-to-speech engine (pyttsx3) that generates clear and timely alerts for detected objects.
- Customizable Target List: A user-defined text file that specifies which objects the system should recognize and notify about, allowing flexibility and personalization.

The combination of these components results in a highly adaptable and accessible system suitable for diverse applications, such as assisting visually impaired users, enhancing robotic navigation, or improving security systems.

1.4 Problem Statement

Despite significant progress in computer vision, there remains a lack of accessible, real-time object detection systems tailored for individuals with specific needs, such as visual impairments. Existing technologies may be limited by factors such as high costs, lack of customization, or insufficient real-time performance. This project addresses these limitations by developing a solution that is:

1. Efficient: Utilizing YOLO for fast and accurate object detection.
2. Accessible: Providing auditory feedback for easy use by visually impaired users.
3. Customizable: Allowing users to define target objects, making it adaptable for various scenarios.

The ultimate goal is to create a tool that not only meets accessibility requirements but also has potential applications in fields such as surveillance, robotics, and autonomous navigation.

1.5 Objective of the Work

The primary objectives of this project are as follows:

1. Real-Time Object Detection: To design a system capable of detecting objects in live video feeds using the YOLO algorithm and OpenCV library.
2. Auditory Feedback Integration: To implement a text-to-speech engine that provides immediate notifications when target objects are detected, ensuring user awareness.
3. Customizable Target Selection: To enable users to specify target objects via a text file, making the system versatile and user-centric.

4. Practical Applications: To create a solution applicable to real-world scenarios, such as assisting visually impaired individuals, enhancing security systems, and supporting robotic vision.

1.6 Organization of the Project

The project is organized into the following chapters:

- Chapter 1: Introduction, including the project's motivation, objectives, and overview of key features.
- Chapter 2: Literature review and related work, discussing the YOLO algorithm, text-to-speech integration, and similar assistive technologies.
- Chapter 3: Methodology, detailing the system architecture, software tools, and implementation process.
- Chapter 4: Results and analysis, presenting the system's performance, use cases, and limitations.
- Chapter 5: Conclusion and future work, summarizing the project's outcomes and potential enhancements.

1.7 Summary

This chapter provided an overview of the project's vision, motivation, and goals. By combining advanced computer vision techniques with auditory feedback, the proposed system seeks to address real-world challenges faced by individuals with visual impairments or those requiring enhanced situational awareness. The integration of YOLO object detection and text-to-speech technology offers a unique solution that is both efficient and user-friendly, with broad applications across various domains.

This work highlights the potential of deep learning and assistive technology to create meaningful and impactful solutions, demonstrating the power of innovation to improve lives and bridge accessibility gaps.

CHAPTER 2: RELATED WORK INVESTIGATION

2.1 Introduction

This chapter explores the existing literature, technologies, and methodologies relevant to the development of real-time object detection and auditory feedback systems. By analyzing the core area of the project, the chapter highlights various approaches used in similar contexts, evaluates their effectiveness, and identifies gaps that the proposed system seeks to address. This investigation provides a foundation for understanding the project's technical direction and its potential to contribute to the field.

2.2 Core Area of the Project

The core area of the project lies at the intersection of computer vision, deep learning, and assistive technology. Specifically, the project focuses on:

1. Object Detection: Utilizing real-time algorithms, particularly YOLO (You Only Look Once), for accurate and efficient object recognition.
2. Auditory Feedback: Converting detected object information into speech using text-to-speech (TTS) systems, enabling accessibility for visually impaired users.
3. Customization: Enabling users to define target objects for personalized and versatile use.
4. Integration: Combining detection and feedback into a cohesive system for real-world applications such as navigation assistance, robotic systems, and enhanced situational awareness.

2.3 Existing Approaches/Methods

2.3.1 Approaches/Methods - 1: Traditional Object Detection

Traditional methods for object detection relied on handcrafted features and algorithms such as:

- Haar Cascades: Used for face detection, Haar cascades identify patterns in grayscale images but struggle with complex scenes and non-frontal perspectives.
- Histogram of Oriented Gradients (HOG): Effective for detecting objects like pedestrians but limited by computational inefficiency and inability to handle overlapping objects.
- Sliding Window Methods: These methods scan the image systematically but are slow and lack scalability for real-time applications.

Advantages: Simplicity and computational efficiency for specific, limited applications.
Disadvantages: Poor accuracy in diverse and complex environments, lack of adaptability to new objects.

2.3.2 Approaches/Methods - 2: Deep Learning-Based Object Detection

With the advent of deep learning, methods like YOLO, SSD (Single Shot MultiBox Detector), and Faster R-CNN have revolutionized object detection by enabling real-time performance with high accuracy.

- YOLO (You Only Look Once): YOLO divides the image into grids and predicts bounding boxes and class probabilities for objects within those grids in a single forward pass. This makes it extremely fast and suitable for real-time applications.
- SSD: Combines high detection speed with multi-scale feature maps but sacrifices some accuracy for smaller objects.
- Faster R-CNN: Offers high accuracy but is computationally intensive, making it less suitable for real-time systems.

Advantages: High accuracy, scalability, and adaptability to diverse environments.
Disadvantages: Computational complexity, reliance on large datasets for training.

2.3.3 Approaches/Methods - 3: Assistive Technologies with Auditory Feedback

Assistive technologies have focused on enhancing accessibility for visually impaired users through sensory substitution.

- Wearable Devices: Devices like OrCam MyEye use cameras and TTS systems to describe surroundings, but they are expensive and limited to specific functions.
- Smartphone Applications: Apps like Seeing AI leverage object detection and speech synthesis to describe scenes, offering portability but depending heavily on internet connectivity and smartphone capabilities.

Advantages: User-centric design, integration of detection and feedback.
Disadvantages: High cost, lack of customization, and dependence on external hardware or connectivity.

2.4 Pros and Cons of the Stated Approaches/Methods

Method	Pros	Cons
Traditional Object Detection	Simple, low computational cost.	Poor performance in complex scenes, low adaptability.
Deep Learning-Based Methods	High accuracy, real-time capabilities, handles diverse environments.	High computational requirements, requires extensive training data.
Assistive Technologies	Enhances accessibility, integrates feedback mechanisms.	Expensive, limited functionality, dependent on additional hardware.

2.5 Issues/Observations from Investigation

1. Efficiency vs. Complexity: While deep learning methods like YOLO provide high accuracy, they require significant computational resources, which may limit their use in low-power devices.
2. Cost Barrier: Existing assistive technologies are often prohibitively expensive, making them inaccessible to many users.
3. Limited Customization: Most solutions do not allow users to define specific objects of interest, reducing their adaptability to unique needs.
4. Integration Challenges: Few systems effectively combine detection and auditory feedback in a seamless manner for real-time use.

2.6 Summary

This chapter reviewed existing methods for object detection and assistive technologies, highlighting their strengths and limitations. Traditional approaches lack the sophistication needed for real-time applications, while deep learning-based methods excel in accuracy but are computationally demanding. Assistive technologies provide valuable functionality but often fall short in terms of cost and customization.

By addressing these gaps, the proposed system aims to deliver an affordable, customizable, and efficient solution that combines real-time object detection with auditory feedback, enhancing accessibility and practicality in real-world scenarios.

CHAPTER 3: REQUIREMENT ARTIFACTS

3.1 Introduction

Requirement analysis is a crucial phase in any project, ensuring that all necessary resources, functionalities, and constraints are clearly identified and documented. This chapter outlines the hardware and software requirements for the project, along with specific project needs such as data, functionality, performance, security, and user interface requirements. These artifacts guide the design and implementation phases, ensuring the system meets its intended objectives efficiently and effectively.

3.2 Hardware and Software Requirements

3.2.1 Hardware Requirements

The system requires the following hardware components for effective functionality:

- Computer/Processor: A system with at least 8 GB RAM and a multi-core processor (Intel i5 or equivalent) for handling YOLO's computational demands.
- Web Camera: A high-resolution webcam (720p or above) for real-time video capture.
- GPU (Optional): For enhanced performance, a GPU (e.g., NVIDIA GTX 1050 or higher) can be utilized to speed up YOLO processing.
- Microphone and Speakers: Necessary for auditory feedback using the text-to-speech engine.

3.2.2 Software Requirements

- Operating System: Windows 10, macOS, or Linux.
- Programming Language: Python 3.x.
- Libraries and Frameworks:
    - OpenCV: For image and video processing.
    - NumPy: For efficient numerical operations.
    - pyttsx3: For text-to-speech conversion.
    - YOLO model files (yolov3.weights, yolov3.cfg, and coco.names): For object detection.
- Development Environment: Any IDE or text editor like PyCharm, Visual Studio Code, or Jupyter Notebook.

3.3 Specific Project Requirements

3.3.1 Data Requirement

- Input Data: Live video feed from the webcam serves as the primary data source.
- Model Data: Pre-trained YOLO model files (yolov3.weights and yolov3.cfg) and the coco.names file containing the class labels for recognized objects.
- Target Object List: A text file (target_objects.txt) specifying the user-defined objects to be detected and announced.

3.3.2 Functional Requirements

- Real-Time Object Detection:
    - The system must capture video frames and detect objects using the YOLO algorithm.
    - Objects should be identified within a confidence threshold (e.g., > 50%).
- Auditory Feedback:
    - The system must announce detected target objects using the pyttsx3 library.
- Customizable Targets:
    - Users must be able to edit the target_objects.txt file to define or modify the objects they wish to detect.
- User Controls:
    - The system should provide a mechanism to start, pause, or stop the detection process.

3.3.3 Performance and Security Requirements

- Performance Requirements:
    - The system should process at least 15-30 frames per second to ensure real-time detection.
    - Detection latency should be minimal to provide instantaneous feedback.
- Security Requirements:
    - The application should handle sensitive data (if any) securely and ensure that live video feeds are not stored or shared without user consent.
    - Proper exception handling must be implemented to prevent crashes due to invalid inputs or hardware issues.

3.3.4 Look and Feel Requirements

- User Interface:

    - The application should provide a simple and intuitive interface with minimal setup requirements.
    - The video feed should display bounding boxes around detected objects with labels for clear visualization.
- Visual Cues:
    - Detected objects should be highlighted with distinct colors, and their labels should be legible, even in varying lighting conditions.
- Auditory Clarity:
    - The text-to-speech system should use a clear and understandable voice, customizable for speed and tone if possible.

3.3.5 Additional Requirements

- System Scalability: The architecture should support potential future enhancements, such as adding more object classes or integrating additional feedback mechanisms like vibrations or visual alerts.
- Offline Functionality: The system must operate offline without requiring an active internet connection for model processing or text-to-speech synthesis.

3.4 Summary

This chapter outlined the key requirements for the project, detailing the hardware and software components, data inputs, functional capabilities, and design considerations. By addressing these requirements, the project ensures a robust and user-friendly solution that integrates real-time object detection with auditory feedback. These specifications serve as the foundation for implementing a system that is accessible, efficient, and adaptable to user needs.

CHAPTER 4: DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and Goal

The primary goal of this project is to design and implement a real-time object detection and auditory feedback system that assists users, particularly those with visual impairments, in identifying and responding to specific objects in their environment. This is achieved through a structured methodology involving object detection via YOLO, text-to-speech synthesis for feedback, and a user-friendly interface for customization.

The methodology encompasses:

1. Problem Analysis: Identifying user needs, such as real-time performance and auditory clarity.
2. System Design: Creating modular and scalable architecture to facilitate detection and feedback processes.
3. Implementation: Employing pre-trained YOLO models and integrating text-to-speech functionality.
4. Testing and Validation: Ensuring accuracy, speed, and user satisfaction in diverse scenarios.
5. Deployment: Providing a standalone system that works offline and is customizable to user requirements.

The novelty lies in combining state-of-the-art object detection algorithms with customizable auditory feedback, creating a cost-effective, offline, and user-centric solution.

4.2 Functional Modules Design and Analysis

The system is divided into several functional modules:

4.2.1 Input Module

- Function: Captures real-time video feed from the webcam.
- Analysis: Ensures high-resolution input for accurate detection, even in challenging environments such as low light or crowded scenes.

4.2.2 Detection Module

- Function: Utilizes YOLOv3 to detect objects within the video frames.
- Analysis: Optimized for speed and accuracy, processing up to 30 frames per second while maintaining detection precision.

4.2.3 Feedback Module

- Function: Converts detected object names into auditory feedback using the pyttsx3 library.
- Analysis: Configurable speech speed and tone ensure clear and user-friendly feedback.

4.2.4 Customization Module

- Function: Reads a user-defined list of target objects from a file (target_objects.txt).
- Analysis: Empowers users to specify their unique requirements, enhancing adaptability across various applications.

4.2.5 Output Module

- Function: Displays detected objects with bounding boxes and labels on the video feed.
- Analysis: Ensures visual feedback aligns with auditory cues for enhanced usability.

4.3 Software Architectural Designs

The system follows a modular and layered architecture:

1. Input Layer: Handles video capture and preprocesses frames for the detection module.
2. Processing Layer:
    - YOLO-based detection for identifying objects.
    - Non-Maximum Suppression (NMS) to refine bounding boxes and eliminate redundancies.
3. Feedback Layer: Generates auditory feedback corresponding to the detected objects.
4. User Interaction Layer: Provides an interface for customization and displays results visually and aurally.

Data Flow Diagram:

- Input → YOLO Processing → Bounding Box Generation → Auditory Feedback → Output Display

4.4 Subsystem Services

Each subsystem is designed to handle specific tasks efficiently:

1. Video Processing Subsystem:
    - Reads frames from the webcam and prepares them for analysis.
2. Detection Subsystem:
    - Runs the YOLO algorithm, identifies objects, and computes confidence scores.

3. Feedback Subsystem:
    - Matches detected objects with the user-defined target list and generates corresponding speech.
4. Interface Subsystem:
    - Provides options for customization and displays detected objects visually.

4.5 User Interface Designs

The user interface focuses on simplicity and accessibility:

4.5.1 Video Feed Display

- Real-time visualization of the webcam feed with bounding boxes and labels for detected objects.
- Green bounding boxes highlight target objects.

4.5.2 Customization Panel

- A simple text file (target_objects.txt) allows users to add or remove target objects.

4.5.3 Control Mechanisms

- Keyboard shortcuts for starting/stopping detection (q to quit).
- Feedback speed and tone adjustment options via a configuration file or UI sliders (future enhancement).

4.5.4 Visual Aesthetic

- Clean, minimalistic interface ensuring focus remains on the detection results.
- High contrast colors for bounding boxes and labels to enhance visibility.

4.6 Summary

This chapter detailed the design methodology, emphasizing the modular architecture and novel integration of object detection and auditory feedback for assistive technology. By systematically addressing input, processing, feedback, and user interaction, the design ensures a robust and user-friendly system. The proposed methodology's novelty lies in its customizable, cost-effective, and offline-capable nature, making it a valuable tool for diverse applications.

CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

7.1 Outline

This chapter provides a comprehensive summary of the project's accomplishments, addressing its limitations, constraints, and areas for improvement. Recommendations for future enhancements are discussed to ensure the system's continuous evolution. The inference highlights the overall impact of the project on its intended users and applications.

7.2 Limitations/Constraints of the System

Despite its success, the system has a few limitations and constraints:

1. Detection Range and Accuracy:
    - The system's accuracy decreases for small objects or objects located at a significant distance from the camera.
2. Lighting Conditions:
    - Performance is affected by poor lighting or excessive glare, which can hinder object detection.
3. Hardware Dependence:
    - Real-time processing requires moderate to high computational resources, such as a dedicated GPU for optimal performance.
4. Limited Customization Options:
    - Currently, users can only modify the target objects list through a text file. A graphical interface for customization is not yet implemented.
5. Object Overlap:
    - Overlapping or occluded objects may reduce detection accuracy, particularly in crowded environments.

7.3 Future Enhancements

To address the identified limitations and expand the system's capabilities, the following future enhancements are recommended:

1. Improved Detection Algorithms:
    - Upgrade to YOLOv5 or YOLOv8 for improved accuracy, speed, and efficiency.
2. Dynamic Lighting Adjustments:

    - Integrate preprocessing techniques, such as histogram equalization, to improve detection in challenging lighting conditions.

3. Cross-Platform Compatibility:
    - Develop a mobile-friendly version of the system to make it more accessible to users.
4. Advanced Customization Interface:
    - Implement a graphical user interface (GUI) for easier customization of target objects and system settings.
5. Multimodal Feedback:
    - Incorporate haptic or visual alerts to complement auditory feedback, enhancing usability for diverse user groups.
6. Integration with IoT:
    - Connect the system to IoT platforms for broader applications, such as automated alerts in smart homes or warehouses.
7. Expanded Training Dataset:
    - Train the model with additional classes of objects to increase its applicability across various domains.
8. Energy Efficiency Optimization:
    - Optimize the system to run on low-power devices without compromising performance.

7.4 Inference

This project successfully developed a real-time object detection system with auditory feedback, providing a practical tool for assistive technology and other applications. Its limitations are outweighed by its adaptability, offline capability, and ease of use. The proposed future enhancements aim to refine the system further, expanding its functionality, accessibility, and scope.

In conclusion, this project demonstrates how advanced AI techniques, such as YOLO and text-to-speech integration, can create impactful and innovative solutions. With continued development, the system holds the potential to become a powerful tool in diverse fields, including assistive technology, security, and robotics.

Appendix A: Additional Information

A.1 System Configuration

- *Hardware Requirements:*
  - CPU: Intel i5 or higher
  - RAM: 8GB or higher
  - GPU: NVIDIA GTX 1060 or equivalent (for real-time performance)
  - Webcam: 720p or higher resolution
- *Software Requirements:*
  - Python 3.x
  - OpenCV library (for computer vision)
  - pyttsx3 (for text-to-speech synthesis)
  - YOLOv3 pre-trained weights, config, and class files (coco.names)
  - Operating System: Windows, macOS, or Linux

A.2 Installation Instructions

1. Install the necessary libraries:
   - pip install opencv-python pyttsx3
2. Download the YOLOv3 weights, configuration, and coco.names file from official repositories.
3. Ensure the target_objects.txt file is populated with the objects the user wishes to detect.
4. Run the Python script and start detecting objects in real-time via the webcam.

Appendix B: Additional Code and Resources

B.1 Algorithms and Flowcharts

To make it easier for users or reviewers to understand the logic and flow of the system, you can add diagrams, flowcharts, and algorithm descriptions. For example:

1. Flowchart of Object Detection Process
    - This would visually represent the steps in detecting and processing objects in a video frame, from loading the YOLO model to applying non-maxima suppression.
2. Algorithmic Steps
    - Outline the steps of the main object detection algorithm, such as:
        1. Input: Capture video frame
        2. Preprocessing: Resize and normalize image for YOLO
        3. Object Detection: YOLO model detects objects in the frame
        4. Postprocessing: Apply non-maxima suppression
        5. Output: Draw bounding boxes and give audio feedback

This breakdown can help users understand the technical flow behind the detection process.

B.2 Dataset Description

Provide details about the dataset used, such as the COCO dataset (for YOLOv3) and how it relates to the detected objects. You can also mention how the dataset is processed or any custom dataset that was used for training or testing.

Example:

- Dataset: The system uses the COCO (Common Objects in Context) dataset, which contains a wide variety of objects in real-world scenarios. The pre-trained YOLOv3 model leverages this dataset for object recognition.

B.3 Configuration Files

You can provide snippets of the configuration files that are used in the project, such as:

1. coco.names
    - A text file listing the object classes YOLO can recognize, like:

python

Copy code

person

bicycle

car

dog

cat

...

2. target_objects.txt
    - This file allows users to list the specific objects they want the system to focus on. Example:

Copy code

person

dog

laptop

B.4 Troubleshooting and Known Issues

Provide helpful troubleshooting tips and solutions to common issues users may encounter while setting up or running the system. You can also mention any known bugs or issues and how they might be resolved in future versions.

Example:

- Issue 1: Low detection accuracy in poor lighting conditions.
  Solution: Consider using better lighting or enabling a preprocessing step like histogram equalization to adjust the image's brightness and contrast.
- Issue 2: Model running too slow on lower-end systems.
  Solution: Use a smaller YOLO model (e.g., YOLO-tiny) for faster performance, or ensure the system meets the required GPU specifications.

B.5 External Libraries and Tools

You can list the libraries and tools that the project depends on and provide brief descriptions of their functionality.

Example:

- OpenCV: An open-source computer vision library used for image processing, object detection, and video capture.
- pyttsx3: A Python library used for text-to-speech conversion that gives audio feedback when objects are detected.
- YOLOv3: A state-of-the-art object detection algorithm that performs real-time detection with high accuracy.

B.6 Example Output

You could include some example outputs (images or text) to show how the system works in real-time. This could include screenshots of the object detection in action, showing bounding boxes drawn on detected objects, or logs of detected objects and their confidence scores.

Example Output:

- Detected Objects:
  person (Confidence: 0.85)
  dog (Confidence: 0.92)
  laptop (Confidence: 0.76)

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